

Disproof of Conjecture 00000004316

Two local-density interpretations

gaochengzhecpu

Source and scope. The conjecture calls local density the density of quadratic-form representability at a prime; the Chinese wording specifically refers to existence of representations. It asserts a universal product value $1/2$, but specifies neither a particular form nor a target, and gives no positivity or local-solubility restriction. We first treat the proportion of right-hand sides that have a representation. We then separately treat the customary normalized number of representations of a fixed target. These are different densities and are not interchanged.

1. Existence density: every target is represented

Let $Q(x, y) = xy$ over $\mathbb{Z}$. This is an integral, nondegenerate quadratic form. Its integral polar bilinear form is

$$B_Q((x, y), (u, v)) = Q(x + u, y + v) - Q(x, y) - Q(u, v) = xv + uy,$$

with matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and determinant -1 . Thus it is not a degenerate zero form. It is indefinite, which is allowed because the source imposes no positive-definiteness condition.

For $q \geq 1$, define the represented residue set and its proportion by

$$E_Q(q) = \{a \in \mathbb{Z}/q\mathbb{Z} : \exists x, y \in \mathbb{Z}/q\mathbb{Z}, xy = a\}, \quad d_Q(q) = \frac{\#E_Q(q)}{q}.$$

Theorem 1. *Every local existence density for $Q(x, y) = xy$ equals 1, and its prime-ordered product exists and equals 1, not $1/2$.*

Proof. For every residue class a , the pair $(a, 1)$ gives $Q(a, 1) = a$. Hence $E_Q(q) = \mathbb{Z}/q\mathbb{Z}$, its actual cardinality is q , and $d_Q(q) = 1$. For every prime p this gives

$$\delta_p = \lim_{h \rightarrow \infty} d_Q(p^h) = 1, \quad \prod_{\substack{p \leq N \\ p \text{ prime}}} \delta_p = 1.$$

The empty initial product is also 1. Thus the product converges to $1 \neq 1/2$. □

This main counterexample directly addresses existence for a varying right-hand side. All its local factors genuinely exist. It does not rely on an obstructed target or an undefined product.

2. Fixed-target density: a zero local factor

For the separate, customary fixed-target interpretation, take the positive-definite nondegenerate integral form $F(x, y) = x^2 + y^2$ and target $t = 3$. Define

$$R(q) = \#\{(x, y) \in (\mathbb{Z}/q\mathbb{Z})^2 : x^2 + y^2 \equiv 3 \pmod{q}\}, \quad \sigma_p = \lim_{h \rightarrow \infty} \frac{R(p^h)}{p^h},$$

when the latter limit exists. Here the count is of representing pairs for one target, not of represented right-hand sides.

Every square modulo 4 is 0 or 1, so two squares cannot sum to 3 modulo 4. Reduction modulo 4 shows $R(q) = 0$ whenever $4 \mid q$. Consequently

$$R(2^h) = 0 \quad (h \geq 2), \quad \sigma_2 = 0.$$

Theorem 2. *There is no family of real fixed-target local-density limits for $F(x, y) = 3$ whose prime-ordered product converges to $1/2$.*

Proof. For any such family $(L_p)_p$, uniqueness of the limit at 2 forces $L_2 = 0$. Every partial product over primes $p \leq N$ is therefore zero for $N \geq 2$. These products tend to zero, excluding the limit $1/2$ by uniqueness. $\square$

This negates the combined existence-and-value assertion without assuming convergence of the other local factors. The source does not restrict fixed targets to locally soluble ones. A statement with such a restriction or a different normalization would be different. Neither argument needs to assess the additional convergence-rate claim because the asserted universal value already fails.

Lean correspondence

Both proofs are in one `Main.lean` and one library. The main argument is in namespace `Conjecture4316.ExistenceDensity`. Its `splitForm` is an actual Mathlib integral `QuadraticForm`, with evaluation proved to be x_0x_1 . Its actual polar bilinear form has the stated coordinate matrix and determinant -1 ; its kernel is separately proved trivial.

The finite set `represented q` consists of `Fin q` targets with an existential modular representation. A bridge theorem identifies the counted multiplication with evaluation of the actual form. The witness is the target representative together with $1 \bmod q$, also covering $q = 1$. Lean proves that this set is the full residue universe and has cardinality q . The genuine density is therefore $q/q = 1$ for positive q . All prime-power local limits and all finite prime products are then proved equal to one. The formal proof both constructs a family of existing factors and excludes any consistent family with product limit $1/2$.

The original fixed-target proof is retained in namespace `Conjecture4316`. Its `Q` is the actual form $x_0^2 + x_1^2$. Its `Solutions q` subtype counts canonical residue pairs representing 3, with form evaluation explicitly connected to the count. The modulo-4 obstruction is checked over every residue pair and transferred to every multiple of 4. Lean proves the counts at 2^{e+2} vanish, the factor at 2 is zero, and all sufficiently long prime products vanish. Omitting the initial exponents does not change the local limit.

Reproduction and source

Inside `lean/`, run `lake build`, then

```
lake env lean -DwarningAsError=true Main.lean
```

The project pins Lean 4.19.0, Mathlib, and all transitive revisions. Actual fresh-build logs, standard axiom dependencies, source hashes, compiler results, PDF inspection, and development/review history accompany the submission. No numerical approximation or auxiliary program is required. Compile with `tectonic main.tex`.

The unchanged bilingual statement is included as `SOURCE.md`; see The Last Math Competition, Conjecture 00000004316. Its upstream commit and SHA-256 provenance are recorded in `verification/`.